

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12417615

Kind Code

B2

Date of Patent

September 16, 2025

Inventor(s)

Mukherjee; Maharaj et al.

System and method for consolidation of alerts and events using image matching of heatmap descriptions of infrastructure status

Abstract

A computing platform may train, using historical telemetry state images, an image comparison model to identify matches between telemetry state images. The computing platform may generate a plurality of system alerts corresponding to a period of time. The computing platform may access telemetry data corresponding to the period of time. The computing platform may generate, based on the telemetry data and for a time corresponding to each of the plurality of system alerts, a telemetry state image. The computing platform may input, into the image comparison model, the telemetry state images to identify whether or not any of the plurality of telemetry state images match. Based on detecting a match, the computing platform may consolidate system alerts corresponding to the matching telemetry state images, which may produce a single system alert and may send, to a user device, the single system alert.

Inventors: Mukherjee; Maharaj (Poughkeepsie, NY), Raj; Utkarsh (Charlotte, NC), Murphy; Colin (Charlotte, NC), Benda; Carl (Charlotte, NC), Nyamwange; Elvis (Little Elm, TX), Yarabolu; Vijay Kumar (Hyderabad, IN), Choudhury; Suman Roy (Jersey City, NJ)

Applicant: Bank of America Corporation (Charlotte, NC)

Family ID: 1000008820808

Assignee: Bank of America Corporation (Charlotte, NC)

Appl. No.: 18/197924

Filed: May 16, 2023

Prior Publication Data

Document Identifier

US 20240386696 A1

Publication Date

Nov. 21, 2024

Publication Classification

Int. Cl.: **G06K9/00** (20220101); **G06V10/74** (20220101); **G06V10/82** (20220101); **G08B21/00** (20060101)

U.S. Cl.:

CPC **G06V10/761** (20220101); **G06V10/82** (20220101); **G08B21/00** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11625293	12/2022	Ratkovic et al.	N/A	N/A
2014/0278496	12/2013	Spencer	N/A	N/A
2014/0310564	12/2013	Mallige et al.	N/A	N/A
2015/0149134	12/2014	Mehta et al.	N/A	N/A
2016/0041948	12/2015	Vichare et al.	N/A	N/A
2016/0155315	12/2015	McElhinney	340/502	G06N 5/02
2016/0266962	12/2015	Rajasekharan et al.	N/A	N/A
2018/0157963	12/2017	Salti et al.	N/A	N/A
2019/0139441	12/2018	Akella et al.	N/A	N/A
2019/0179723	12/2018	Allin et al.	N/A	N/A
2019/0199602	12/2018	Zhang et al.	N/A	N/A
2020/0125465	12/2019	Choi et al.	N/A	N/A
2020/0201697	12/2019	Torng et al.	N/A	N/A
2020/0295986	12/2019	Rathinasabapathy et al.	N/A	N/A
2021/0056678	12/2020	Al Shehri et al.	N/A	N/A
2022/0246657	12/2021	Lin et al.	N/A	N/A
2022/0269035	12/2021	Kimura et al.	N/A	N/A
2022/0271072	12/2021	Miyata	N/A	N/A
2022/0295000	12/2021	Yamazaki et al.	N/A	N/A
2022/0295006	12/2021	Kashiwagi	N/A	N/A
2022/0310684	12/2021	Sakata et al.	N/A	N/A
2022/0321791	12/2021	Lin et al.	N/A	N/A
2022/0372181	12/2021	Iwata et al.	N/A	N/A
2022/0392944	12/2021	Tochigi et al.	N/A	N/A
2022/0399469	12/2021	Minari	N/A	N/A
2022/0408046	12/2021	Katou et al.	N/A	N/A
2022/0415947	12/2021	Hasegawa	N/A	N/A
2022/0417464	12/2021	Tatsuzawa	N/A	N/A
2023/0005981	12/2022	Ito et al.	N/A	N/A
2023/0013149	12/2022	Akiyama	N/A	N/A
2023/0014856	12/2022	Goto	N/A	N/A
2023/0020741	12/2022	Liu et al.	N/A	N/A

2023/0058009	12/2022	Tsuchimoto et al.	N/A	N/A
2023/0073737	12/2022	Li et al.	N/A	N/A
2023/0110102	12/2022	Li et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10201610116	12/2017	SG	N/A

OTHER PUBLICATIONS

Dec. 27, 2024—(US) Office Action—U.S. Appl. No. 18/198,367. cited by applicant

Di Liu et al. “An Improved Dynamic Load-balancing Model” IEEE (Year 2016). cited by applicant

Javier Lopez et al. “A Resilient Architecture for the Smart Grid” IEEE (Year 2018). cited by applicant

Emre Gures “Machine Learning-Based Load Balancing Algorithms in Future Heterogeneous Networks: A Survey” IEEE (Year 2022). cited by applicant

Li Liu et al. “Machine Learning-Based Load Balancing Algorithms in Future Heterogeneous Networks: A Survey” IEEE (Year 2007). cited by applicant

Areeg Samir et al. “A Controller Architecture for Anomaly Detection, Root Cause Analysis and Self-Adaptation for Cluster Architectures” Year 2019. cited by applicant

Qing Wang et al. “Constructing the Knowledge Base for Cognitive IT Service Management” (Year 2017). cited by applicant

Jan. 2, 2025—(US) Notice of Allowance—U.S. Appl. No. 18/201,817. cited by applicant

Li Liu et al. “Fault Detection, Diagnostics, and Prognostics: Software Agent Solutions” IEEE Transactions on Vehicular Technology, vol. 56, No. 4 (Year Jul. 2007). cited by applicant

Apr. 8, 2025—(US) Notice of Allowance—U.S. Appl. No. 18/201,817. cited by applicant

Apr. 11, 2025—(US) Office Action—U.S. Appl. No. 18/198,367. cited by applicant

Primary Examiner: Park; Sj

Attorney, Agent or Firm: Banner & Witcoff, Ltd.

Background/Summary

BACKGROUND

(1) In some instances, infrastructure and/or application monitoring tools may generate alerts for a number of reasons. Such tools often generate several alerts that may be related to the same incident or root cause. Because each alert may provoke investigation by a system engineer, it may create inefficiencies if multiple alerts related to the same incident or root cause are generated. Accordingly, it may be important to improve the methods of alert generation for such tools.

SUMMARY OF THE INVENTION

(2) Aspects of the disclosure provide effective, efficient, scalable, and convenient technical solutions that address and overcome the technical problems associated with system alerts. In accordance with one or more embodiments of the disclosure, a computing platform comprising at least one processor, a communication interface, and memory storing computer-readable instructions may train, using historical telemetry state images, an image comparison model to identify matches between telemetry state images. The computing platform may generate, for a system, a plurality of system alerts corresponding to a period of time. The computing platform may access telemetry data corresponding to the period of time. The computing platform may generate, based on the telemetry data and for a time corresponding to each of the plurality of system alerts, a telemetry state image

to produce a plurality of telemetry state images. The computing platform may input, into the image comparison model, the plurality of telemetry state images to identify whether or not any of the plurality of telemetry state images match. Based on detecting a match, the computing platform may: 1) consolidate system alerts corresponding to the matching telemetry state images, which may produce a single system alert representative of the system alerts corresponding to the matching telemetry state images, and 2) send, to a user device, the single system alert and one or more commands directing the user device to display the single system alert, which may cause the user device to display the single system alert.

(3) In one or more instances, training the image comparison model may include training the image comparison model to perform a structural property comparison between telemetry state images to identify whether or not there is a match between the telemetry state images. In one or more instances, training the image comparison model may include training the image comparison model to perform pattern matching using a convolutional neural network (CNN) to identify whether or not there is a match between the telemetry state images.

(4) In one or more examples, identifying whether or not any of the plurality of telemetry state images match may include: 1) identifying a matching score between at least two of the telemetry state images; 2) comparing the matching score to a matching threshold; 3) based on identifying that the matching score meets or exceeds the matching threshold, identifying that the at least two of the telemetry state images match; and 4) based on identifying that the matching score does not meet or exceed the matching threshold, identifying that the at least two of the telemetry state images do not match.

(5) In one or more instances, based on failing to detect the match, the computing platform may send, to the user device, the plurality of system alerts and one or more commands directing the user device to display the plurality of system alerts, which may cause the user device to display the plurality of system alerts. In one or more instances, the computing platform may update, based on identifying whether or not any of the plurality of telemetry state images match, the image comparison model.

(6) In one or more examples, based on detecting the match, the computing platform may send a single alert resolution command directing a network system to execute one or more alert resolution actions to address a system issue noted in the single system alert. In one or more examples, based on failing to detect the match, the computing platform may send a plurality of alert resolution commands directing a network system to execute one or more alert resolution actions to address system issues noted in each of the plurality of system alerts. In one or more examples, the computing platform may normalize the telemetry data, and generating the plurality of telemetry state images may include generating, using the normalized telemetry data, the plurality of telemetry state images.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The present disclosure is illustrated by way of example and is not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

(2) FIGS. 1A and 1B depict an illustrative computing environment for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments.

(3) FIGS. 2A-2C depict an illustrative event sequence for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments.

(4) FIG. 3 depicts an illustrative method for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments.

(5) FIGS. 4 and 5 depict illustrative user interfaces for consolidating alerts using image matching

of heatmap descriptions in accordance with one or more example embodiments.

(6) FIGS. 6 and 7 depict illustrative diagrams for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments.

DETAILED DESCRIPTION

(7) In the following description of various illustrative embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments in which aspects of the disclosure may be practiced. In some instances other embodiments may be utilized, and structural and functional modifications may be made, without departing from the scope of the present disclosure.

(8) It is noted that various connections between elements are discussed in the following description. It is noted that these connections are general and, unless specified otherwise, may be direct or indirect, wired or wireless, and that the specification is not intended to be limiting in this respect.

(9) The following description relates to a system and method for consolidating alerts using image matching of heatmap descriptions of infrastructure status, as is described further below.

(10) Technology infrastructure application monitoring tools as well as infrastructure monitoring tools may generate alerts for many reasons (e.g., when the devices and applications deteriorate in performance, when the device capacity such as CPU, memory, disk space, or the like get overly used or full, any significant diversion from the normal operating conditions, any device or hardware failures, and/or other reasons).

(11) Whatever the reason might be, the system may generate several alerts that might be related to the same incident or root cause. Incident reports from an internal tool may show alerts and incidents that are most likely generated from the same root cause.

(12) Since each alert may be investigated by a system engineer, it may create inefficiencies if several alerts are created from the same root cause. It may be, therefore, important to consolidate similar alerts together so that the engineers can focus on fewer items.

(13) As described below, thermal images may be used that capture the overall health and capacity of the whole infrastructure system. The thermal image may be created by starting with a table of raw telemetry data. The data may be further normalized to convert each cell into a value between zero and one in floating point numbers. The resulting matrix may be a normalized image. Examples of this normalized image can be displayed by appropriate thresholding and associating a color with each of the threshold ranges.

(14) These normalized images may represent the overall health of the system and can be directly attributed and linked to any events, incidents, and consequently any alerts generated. An image matching program may be used to check if the normalized images associated with two alerts are similar. If they are, the alerts may be categorized as the same.

(15) Image similarity matching may be performed for alerts generated around the same time to categorize them as the same alert. All alerts in the same category may be consolidated and presented as one hyper-alert to the user.

(16) The similarity determination of two featureless images may be performed by consecutive low pass filtering, structural similarity matching, and/or other techniques by using image properties such as average, peaks, troughs, center of gravity, moment, spatial frequency, or the like. These and other features are described in greater details below.

(17) FIGS. 1A-1B depict an illustrative computing environment for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments. Referring to FIG. 1A, computing environment **100** may include one or more computer systems. For example, computing environment **100** may include an alert consolidation platform **102**, telemetry information source **103**, and user device **104**.

(18) Alert consolidation platform **102** may include one or more computing devices and/or other computer components (e.g., processors, memories, communication interfaces, or the like). For example, the alert consolidation platform **102** may be configured to generate, update, and/or

otherwise maintain an image comparison model configured to identify matches between state images. In some instances, the image comparison model may use structural similarities and/or constructive low pass filtering to compare the images. Based on the results of the image matching, the alert consolidation platform **102** may be configured to consolidate alerts corresponding to matching images.

(19) Telemetry information source **103** may be or include one or more computing devices (e.g., servers, server blades, or the like) and/or other computer components (e.g., processors, memories, communication interfaces, and/or other components). In some instances, the telemetry information source **103** may be configured to monitor a plurality of individual systems to collect the corresponding telemetry data. In other instances, the telemetry information source **103** may be the source of the telemetry data itself (e.g., producing the telemetry data). Although a single telemetry information source **103** is shown, any number of telemetry information sources **103** may be included in the system architecture without departing from the scope of the disclosure.

(20) User device **104** may be or include one or more devices (e.g., laptop computers, desktop computer, smartphones, tablets, and/or other devices) configured for use in receiving preemptive resolution information from the alert consolidation platform **102**. In some instances, the user device **104** may be configured to display graphical user interfaces (e.g., system alerts, or the like). Any number of such user devices may be used to implement the techniques described herein without departing from the scope of the disclosure.

(21) Computing environment **100** also may include one or more networks, which may interconnect alert consolidation platform **102**, telemetry information source **103**, and user device **104**. For example, computing environment **100** may include a network **101** (which may interconnect, e.g., alert consolidation platform **102**, telemetry information source **103**, and user device **104**).

(22) In one or more arrangements, alert consolidation platform **102**, telemetry information source **103**, and user device **104** may be any type of computing device capable of receiving a user interface, receiving input via the user interface, and communicating the received input to one or more other computing devices. For example, alert consolidation platform **102**, telemetry information source **103**, user device **104**, and/or the other systems included in computing environment **100** may, in some instances, be and/or include server computers, desktop computers, laptop computers, tablet computers, smart phones, or the like that may include one or more processors, memories, communication interfaces, storage devices, and/or other components. As noted above, and as illustrated in greater detail below, any and/or all of alert consolidation platform **102**, telemetry information source **103**, and user device **104** may, in some instances, be special-purpose computing devices configured to perform specific functions.

(23) Referring to FIG. **1B**, alert consolidation platform **102** may include one or more processors **111**, memory **112**, and communication interface **113**. A data bus may interconnect processor **111**, memory **112**, and communication interface **113**. Communication interface **113** may be a network interface configured to support communication between alert consolidation platform **102** and one or more networks (e.g., network **101**, or the like). Memory **112** may include one or more program modules having instructions that when executed by processor **111** cause alert consolidation platform **102** to perform one or more functions described herein and/or one or more databases that may store and/or otherwise maintain information which may be used by such program modules and/or processor **111**. In some instances, the one or more program modules and/or databases may be stored by and/or maintained in different memory units of alert consolidation platform **102** and/or by different computing devices that may form and/or otherwise make up alert consolidation platform **102**. For example, memory **112** may have, host, store, and/or include alert consolidation module **112a**, alert consolidation database **112b**, and machine learning engine **112c**. Alert consolidation module **112a** may have instructions that direct and/or cause alert consolidation platform **102** to execute advanced optimization techniques to generate, apply, and/or otherwise maintain an image comparison model for use in consolidating system alerts. Alert consolidation

database **112b** may store information used by alert consolidation module **112a**, in executing, generating, applying, and/or otherwise maintaining an image comparison model for use in consolidating system alerts and/or in performing other functions. Machine learning engine **112c** may be used to train, deploy, and/or otherwise refine models used to support functionality of the alert consolidation module **112a** through both initial training and one or more dynamic feedback loops, which may, e.g., enable continuous improvement of the alert consolidation platform **102** and further optimize the consolidation of system alerts.

(24) FIGS. 2A-2C depict an illustrative event sequence for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments. Referring to FIG. 2A, at step **201**, the alert consolidation platform **102** may train an image comparison model. For example, the alert consolidation platform **102** may receive historical telemetry data (e.g., from the telemetry information source **103**, and/or otherwise). The alert consolidation platform **102** may normalize the historical telemetry data to create normalized telemetry data values between zero and one (e.g., in floating point numbers). Based on the normalized telemetry data, the alert consolidation platform **102** may generate telemetry state images, similar to the normalized images depicted in FIGS. 6 and 7.

(25) In some instances, the alert consolidation platform **102** may use these normalized images to train a CNN, as the image comparison model, to identify matching telemetry state images. For example, by inputting these normalized images into the CNN, the alert consolidation platform **102** may train the CNN to recognize features in telemetry state images that may enable the CNN to perform image matching between such telemetry state images. In some instances, the alert consolidation platform **102** may train the CNN using unsupervised techniques to categorize the historical images. In other instances, the alert consolidation platform **102** may train the CNN using partially supervised techniques to categorize the historical images.

(26) Additionally or alternatively, the alert consolidation platform **102** may train the image classification model to identify matches using a structural property comparison between various features of the thermal images such as image peaks and troughs (e.g., number of peaks and troughs, total areas of peaks and troughs, or the like), center of gravity, moment, spatial frequency, and/or other features.

(27) In some instances, the image classification model may be trained to generate an image matching score. In instances where the CNN is used, the CNN may be trained to identify similarities between the colors and the locations of such colors in the state images. In these instances, the CNN may be trained to identify a degree to which the colors and their given locations between images match (e.g., on a percentage basis, or the like).

(28) In the case of the featureless recognition, the image classification model may be trained to generate an image matching score for each feature described above (e.g., peaks and troughs, moment, center of gravity, spatial frequency, or the like), and then combine the various feature image matching scores to create an overall image matching score. In these instances, the smaller the discrepancy between the above described features of an input image and a previously classified image, the higher the matching score, and vice versa.

(29) In some instances, these features may be weighted evenly (e.g., overall image matching score = $(0.25 * \text{peaks and troughs score}) + (0.25 * \text{center of gravity score}) + (0.25 * \text{moment score}) + (0.25 * \text{spatial frequency score})$). Alternatively, the features may be weighted differently. In some instances, the features may initially be weighted evenly, and the weighting may be dynamically adjusted over time (e.g., via a dynamic feedback loop) to weight features higher that may be identified as higher indicators of matching images.

(30) In some instances, the alert consolidation platform **102** may train the image comparison model to identify non-exact (e.g., fuzzy) matches based on a certain percentage of matching thermal image features (e.g., despite an exact match not being available). For example, the alert consolidation platform **102** may predict the fuzzy match in the event that an exact match is not

identified. In some instances, the alert consolidation platform **102** may generate a similarity score between various features of the input thermal images and the historical thermal images. If the similarity score exceeds a predetermined similarity threshold, the alert consolidation platform **102** may identify a fuzzy match. In these instances, if a corresponding match is ultimately identified through a fuzzy match, the alert consolidation platform **102** may train the image comparison model to identify a correlation between the corresponding state images (e.g., by refining the model using a dynamic feedback loop). In doing so, the alert consolidation platform **102** may conserve computing resources by avoiding an extensive alternative evaluation to identify outputs where no exact match is identified.

(31) In some instances, in training the image comparison model, the alert consolidation platform **102** may train a supervised learning model. For example, the alert consolidation platform **102** may train one or more of: decision trees, ensembles (e.g., boosting, bagging, random forest, or the like), neural networks, linear regression models, artificial neural networks, logistic regression models, support vector machines, and/or other supervised learning models. In some instances, the alert consolidation platform **102** may train the image comparison model using one or more unsupervised learning techniques (e.g., classification, regression, clustering, anomaly detection, artificial neural networks, and/or other supervised models/techniques). Accordingly, the image comparison model may ultimately be trained to identify matching state images based on their similarities.

(32) With further reference to FIG. 2A, at step **202**, the alert consolidation platform **102** may generate system alerts for a given system infrastructure being monitored. In some instances, the alerts may correspond to telemetry information from the telemetry information source **103**. For example, the alert consolidation platform **102** may generate system alerts indicating a particular process is missing, that memory utilization dropped below or exceeded various thresholds, that CPU thresholds are not met or are exceeded, that processes are not running, and/or other information. In some instances, these alerts may be generated for incidents identified as occurring within a predetermined period of time of each other.

(33) At step **203**, the telemetry information source **103** may establish a connection with the alert consolidation platform **102**. For example, the telemetry information source **103** may establish a first wireless data connection with the alert consolidation platform **102** to link the telemetry information source **103** to the alert consolidation platform **102** (e.g., in preparation for sending telemetry information). In some instances, the telemetry information source **103** may identify whether or not a connection is already established with the alert consolidation platform **102**. If a connection is already established with the alert consolidation platform **102**, the telemetry information source **103** might not re-establish the connection. If a connection is not yet established with the alert consolidation platform **102**, the telemetry information source **103** may establish the first wireless data connection as described herein.

(34) At step **204**, the alert consolidation platform **102** may access telemetry data from the telemetry information source. For example, the telemetry information source **103** may send time stamps, dates, system names, computer processing unit (CPU) information, memory information, and/or other telemetry information corresponding to performance of a plurality of systems (and/or the telemetry information source **103** itself). In some instances, in accessing the telemetry data, the alert consolidation platform **102** may access telemetry data corresponding to the time period during which the alerts/incidents (e.g., described at step **202**) were generated and/or otherwise identified. In some instances, the alert consolidation platform **102** may access the telemetry data via the communication interface **113** and while the first wireless data connection is established.

(35) Referring to FIG. 2B, at step **205**, the alert consolidation platform **102** may normalize the telemetry data received at step **204**. For example, the alert consolidation platform **102** may convert the telemetry data (which may, e.g., include values of different sizes, ranges, or the like) to values between zero and one. In doing so, the alert consolidation platform **102** may configure the telemetry data for representation as telemetry state images.

(36) At step **206**, the alert consolidation platform **102** may generate telemetry state images using the normalized telemetry data. For example, the alert consolidation platform **102** may generate images similar to diagram **600** depicted in FIG. **6**, the diagram **700** depicted in FIG. **7**, or the like. For example, the telemetry state images may include the telemetry data plotted against the various systems corresponding to the telemetry data and at a given time. Specifically, the telemetry state images may represent heatmaps corresponding to a current status of a system or system infrastructure represented by the telemetry data. In essence, the telemetry state images may be snapshot representations of the performance of these systems at various times. For example, the telemetry state images may each represent a state of the system infrastructure at a time corresponding to a particular alert (e.g., the alerts generated at step **202**).

(37) In some instances, in generating the telemetry state images, the alert consolidation platform **102** may apply one or more thresholding techniques. As a simple example, the alert consolidation platform **102** may use green to represent any values from 0-3 (inclusive), yellow to represent any values from 3.1-6 (inclusive), and red to represent any values from 6.1-10 (inclusive). Any number of colors and/or threshold ranges may be implemented without departing from the scope of the disclosure.

(38) At step **207**, the alert consolidation platform **102** may input the state images, generated at step **206**, into the image comparison model to identify whether there are any matches. For example, the alert consolidation platform **102** may compare each state image to the remaining state images generated at step **206**.

(39) In some instances, the alert consolidation platform **102** may identify matches between the telemetry state images using the CNN. For example, the alert consolidation platform **102** may input the telemetry state images into the CNN, which may, e.g., identify matches by comparing features of the state images. In some instances, the CNN may identify an exact match. In other instances, the CNN may identify a threshold match (e.g., at least a threshold level match).

(40) In some instances, in identifying matches in the telemetry state images, the CNN may receive images in a spatial domain, and may convert (e.g., using a first Fourier transform or otherwise) the images into the frequency domain. In doing so, the CNN may make translations, rotations, inversions, and/or other features of the images invariant, which may, e.g., increase both a speed and an accuracy at which the CNN may classify the images.

(41) Additionally or alternatively, the image comparison model may compare features (e.g., peaks and troughs, center of gravity, moment spatial frequency, and/or other features) of the state images to each other. If the image comparison model identifies that a similarity or matching score with one or more other state images exceeds a predetermined matching threshold, the image comparison model may identify a match between the corresponding state images.

(42) At step **208**, the alert consolidation platform **102** may configure the alerts, generated at step **202**, based on any matches identified using the image comparison model. For example, if the alert consolidation platform **102** identified that a first state image (corresponding to a first alert) matches a second state image (corresponding to a second alert), the alert consolidation platform **102** may combine/consolidate the first and second alerts into a single alert. If instead the alert consolidation platform **102** did not identify any matches between state images, the alert consolidation platform **102** may maintain a configuration of the alerts (e.g., maintain separate first and second alerts, or the like).

(43) Referring to FIG. **2C**, at step **209**, the alert consolidation platform **102** may establish a connection with the user device **104**. For example, the alert consolidation platform **102** may establish a second wireless data connection with the user device **104** to link the alert consolidation platform **102** to the user device **104** (e.g., in preparation for sending alerts). In some instances, the alert consolidation platform **102** may identify whether or not a connection is already established with the user device **104**. If a connection is already established with the user device **104**, the alert consolidation platform **102** might not re-establish the connection. If a connection is not yet

established with the user device **104**, the alert consolidation platform **102** may establish the second wireless data connection as described herein.

(44) At step **210**, the alert consolidation platform **102** may send the system alert(s) to the user device **104**. For example, the alert consolidation platform **102** may send the system alerts to the user device **104** via the communication interface and while the second wireless data connection is established. In some instances, the alert consolidation platform **102** may also send one or more commands directing the user device **104** to display the system alerts.

(45) At step **211**, the user device **104** may receive the system alert(s) sent at step **210**. For example, the user device **104** may receive the system alerts while the second wireless data connection is established. In some instances, the user device **104** may also receive the one or more commands directing the user device **104** to display the system alerts.

(46) At step **212**, based on or in response to the one or more commands received at step **211**, the user device **104** may display the system alerts. For example, the user device **104** may display a graphical user interface similar to graphical user interface **400**, shown in FIG. **4**, graphical user interface **500**, shown in FIG. **5**, or the like.

(47) In addition or as an alternative to sending the system alerts and display commands, the alert consolidation platform **102** may send one or more alert resolution commands, which may, for example, direct the user device **104**, telemetry information source **103**, and/or other systems (such as a network gateway, packet routing system, load balancer, or the like) to automatically execute one or more actions to address the given alerts (which may, e.g., cause these systems to execute the one or more alert resolution actions accordingly). In some instances, such commands may be consolidated as described above with regard to the alerts. For example, if two alerts have been consolidated, the alert consolidation platform **102** may send a single alert resolution command corresponding to the consolidated alert. In contrast, if the two alerts have not been consolidated, the alert consolidation platform **102** may send alert resolution commands corresponding to each alert. In some instances, the alert resolution commands may cause requests to be directed away from a particular system, update processes, and/or otherwise address issues flagged by the alerts. In doing so, processing resources may be conserved by avoiding the duplicate performance of particular actions.

(48) At step **213**, the alert consolidation platform **102** may update the image comparison model. For example, the alert consolidation platform **102** may update the image comparison model based on the telemetry state images and the comparison results. In doing so, the alert consolidation platform **102** may continue to refine the image comparison model using a dynamic feedback loop, which may, e.g., increase the accuracy and effectiveness of the model in performing image matching and alert consolidation.

(49) In some instances, the alert consolidation platform **102** may continuously refine the image comparison model. In some instances, the alert consolidation platform **102** may maintain an accuracy threshold for the image comparison model, and may pause refinement (through the dynamic feedback loops) of the model if the corresponding accuracy is identified as greater than the corresponding accuracy threshold. Similarly, if the accuracy fails to be equal or less than the given accuracy threshold, the alert consolidation platform **102** may resume refinement of the model through the corresponding dynamic feedback loop.

(50) Although the analysis of system telemetry data is primarily described, the methods described above may be used to analyze other types of information (e.g., application performance information, or the like) without departing from the scope of the disclosure.

(51) FIG. **3** depicts an illustrative method for consolidating alerts using image matching of heatmap descriptions in accordance with one or more example embodiments. Referring to FIG. **3**, at step **305**, a computing platform comprising one or more processors, memory, and a communication interface may train an image comparison model. At step **310**, the computing platform may generate system alerts for a system infrastructure. At step **315**, the computing platform may access telemetry

information corresponding to the system alerts. At step **320**, the computing platform may normalize the telemetry information. At step **325**, the computing platform may generate state images using the telemetry information. At step **330**, the computing platform may identify whether or not any matching images are identified.

(52) If no matching images are identified, the computing platform may proceed to step **340**. Otherwise, if matching images are identified, the computing platform may proceed to step **335**. At step **335**, the computing platform may consolidate the corresponding alerts. At step **340**, the computing platform may send the alerts to a user device for display. At step **345**, the computing platform may update the image comparison model.

(53) One or more aspects of the disclosure may be embodied in computer-usable data or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices to perform the operations described herein. Generally, program modules include routines, programs, objects, components, data structures, and the like that perform particular tasks or implement particular abstract data types when executed by one or more processors in a computer or other data processing device. The computer-executable instructions may be stored as computer-readable instructions on a computer-readable medium such as a hard disk, optical disk, removable storage media, solid-state memory, RAM, and the like. The functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents, such as integrated circuits, application-specific integrated circuits (ASICs), field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects of the disclosure, and such data structures are contemplated to be within the scope of computer executable instructions and computer-usable data described herein.

(54) Various aspects described herein may be embodied as a method, an apparatus, or as one or more computer-readable media storing computer-executable instructions. Accordingly, those aspects may take the form of an entirely hardware embodiment, an entirely software embodiment, an entirely firmware embodiment, or an embodiment combining software, hardware, and firmware aspects in any combination. In addition, various signals representing data or events as described herein may be transferred between a source and a destination in the form of light or electromagnetic waves traveling through signal-conducting media such as metal wires, optical fibers, or wireless transmission media (e.g., air or space). In general, the one or more computer-readable media may be and/or include one or more non-transitory computer-readable media.

(55) As described herein, the various methods and acts may be operative across one or more computing servers and one or more networks. The functionality may be distributed in any manner, or may be located in a single computing device (e.g., a server, a client computer, and the like). For example, in alternative embodiments, one or more of the computing platforms discussed above may be combined into a single computing platform, and the various functions of each computing platform may be performed by the single computing platform. In such arrangements, any and/or all of the above-discussed communications between computing platforms may correspond to data being accessed, moved, modified, updated, and/or otherwise used by the single computing platform. Additionally or alternatively, one or more of the computing platforms discussed above may be implemented in one or more virtual machines that are provided by one or more physical computing devices. In such arrangements, the various functions of each computing platform may be performed by the one or more virtual machines, and any and/or all of the above-discussed communications between computing platforms may correspond to data being accessed, moved, modified, updated, and/or otherwise used by the one or more virtual machines.

(56) Aspects of the disclosure have been described in terms of illustrative embodiments thereof. Numerous other embodiments, modifications, and variations within the scope and spirit of the appended claims will occur to persons of ordinary skill in the art from a review of this disclosure.

For example, one or more of the steps depicted in the illustrative figures may be performed in other than the recited order, and one or more depicted steps may be optional in accordance with aspects of the disclosure.

Claims

1. A computing platform comprising: at least one processor; a communication interface communicatively coupled to the at least one processor; and memory storing computer-readable instructions that, when executed by the at least one processor, cause the computing platform to: train, using historical telemetry state images, an image comparison model to identify matches between telemetry state images; generate, for a system, a plurality of system alerts corresponding to a period of time; access telemetry data corresponding to the period of time; generate, based on the telemetry data and for a time corresponding to each of the plurality of system alerts, a telemetry state image to produce a plurality of telemetry state images; input, into the image comparison model, the plurality of telemetry state images to identify whether or not any of the plurality of telemetry state images match; and based on detecting a match: consolidate system alerts corresponding to the matching telemetry state images, wherein consolidating the system alerts corresponding to the matching telemetry state images produces a single system alert representative of the system alerts corresponding to the matching telemetry state images, and send, to a user device, the single system alert and one or more commands directing the user device to display the single system alert, wherein sending the one or more commands directing the user device to display the single system alert causes the user device to display the single system alert.
2. The computing platform of claim 1, wherein training the image comparison model comprises training the image comparison model to perform a structural property comparison between telemetry state images to identify whether or not there is a match between the telemetry state images.
3. The computing platform of claim 1, wherein training the image comparison model comprises training the image comparison model to perform pattern matching using a convolutional neural network (CNN) to identify whether or not there is a match between the telemetry state images.
4. The computing platform of claim 1, wherein identifying whether or not any of the plurality of telemetry state images match comprises: identifying a matching score between at least two of the telemetry state images; comparing the matching score to a matching threshold; based on identifying that the matching score meets or exceeds the matching threshold, identifying that the at least two of the telemetry state images match; and based on identifying that the matching score does not meet or exceed the matching threshold, identifying that the at least two of the telemetry state images do not match.
5. The computing platform of claim 1, wherein the memory stores additional computer readable instructions that, when executed by the at least one processor, cause the computing platform to: based on failing to detect the match, send, to the user device, the plurality of system alerts and one or more commands directing the user device to display the plurality of system alerts, wherein sending the one or more commands directing the user device to display the plurality of system alerts causes the user device to display the plurality of system alerts.
6. The computing platform of claim 1, wherein the memory stores additional computer readable instructions that, when executed by the at least one processor, cause the computing platform to: update, based on identifying whether or not any of the plurality of telemetry state images match, the image comparison model.
7. The computing platform of claim 1, wherein the memory stores additional computer readable instructions that, when executed by the at least one processor, cause the computing platform to: based on detecting the match: send a single alert resolution command directing a network system to execute one or more alert resolution actions to address a system issue noted in the single system

alert.

8. The computing platform of claim 1, wherein the memory stores additional computer readable instructions that, when executed by the at least one processor, cause the computing platform to: based on failing to detect the match, send a plurality of alert resolution commands directing a network system to execute one or more alert resolution actions to address system issues noted in each of the plurality of system alerts.

9. The computing platform of claim 1, wherein the memory stores additional computer readable instructions that, when executed by the at least one processor, cause the computing platform to: normalize the telemetry data, wherein generating the plurality of telemetry state images comprises generating, using the normalized telemetry data, the plurality of telemetry state images.

10. A method comprising: at a computing platform comprising at least one processor, a communication interface, and memory: training, using historical telemetry state images, an image comparison model to identify matches between telemetry state images; generating, for a system, a plurality of system alerts corresponding to a period of time; accessing telemetry data corresponding to the period of time; generating, based on the telemetry data and for a time corresponding to each of the plurality of system alerts, a telemetry state image to produce a plurality of telemetry state images; inputting, into the image comparison model, the plurality of telemetry state images to identify whether or not any of the plurality of telemetry state images match; and based on detecting a match: consolidating system alerts corresponding to the matching telemetry state images, wherein consolidating the system alerts corresponding to the matching telemetry state images produces a single system alert representative of the system alerts corresponding to the matching telemetry state images, and sending, to a user device, the single system alert and one or more commands directing the user device to display the single system alert, wherein sending the one or more commands directing the user device to display the single system alert causes the user device to display the single system alert.

11. The method of claim 10, wherein training the image comparison model comprises training the image comparison model to perform a structural property comparison between telemetry state images to identify whether or not there is a match between the telemetry state images.

12. The method of claim 10, wherein training the image comparison model comprises training the image comparison model to perform pattern matching using a convolutional neural network (CNN) to identify whether or not there is a match between the telemetry state images.

13. The method of claim 10, wherein identifying whether or not any of the plurality of telemetry state images match comprises: identifying a matching score between at least two of the telemetry state images; comparing the matching score to a matching threshold; based on identifying that the matching score meets or exceeds the matching threshold, identifying that the at least two of the telemetry state images match; and based on identifying that the matching score does not meet or exceed the matching threshold, identifying that the at least two of the telemetry state images do not match.

14. The method of claim 10, further comprising: based on failing to detect the match, sending, to the user device, the plurality of system alerts and one or more commands directing the user device to display the plurality of system alerts, wherein sending the one or more commands directing the user device to display the plurality of system alerts causes the user device to display the plurality of system alerts.

15. The method of claim 10, further comprising: updating, based on identifying whether or not any of the plurality of telemetry state images match, the image comparison model.

16. The method of claim 10, further comprising: based on detecting the match: sending a single alert resolution command directing a network system to execute one or more alert resolution actions to address a system issue noted in the single system alert.

17. The method of claim 10, further comprising: based on failing to detect the match, sending a plurality of alert resolution commands directing a network system to execute one or more alert

resolution actions to address system issues noted in each of the plurality of system alerts.

18. The method of claim 10, further comprising: normalize the telemetry data, wherein generating the plurality of telemetry state images comprises generating, using the normalized telemetry data, the plurality of telemetry state images.

19. One or more non-transitory computer-readable media storing instructions that, when executed by a computing platform comprising at least one processor, a communication interface, and memory, cause the computing platform to: train, using historical telemetry state images, an image comparison model to identify matches between telemetry state images; generate, for a system, a plurality of system alerts corresponding to a period of time; access telemetry data corresponding to the period of time; generate, based on the telemetry data and for a time corresponding to each of the plurality of system alerts, a telemetry state image to produce a plurality of telemetry state images; input, into the image comparison model, the plurality of telemetry state images to identify whether or not any of the plurality of telemetry state images match; and based on detecting a match: consolidate system alerts corresponding to the matching telemetry state images, wherein consolidating the system alerts corresponding to the matching telemetry state images produces a single system alert representative of the system alerts corresponding to the matching telemetry state images, and send, to a user device, the single system alert and one or more commands directing the user device to display the single system alert, wherein sending the one or more commands directing the user device to display the single system alert causes the user device to display the single system alert.

20. The one or more non-transitory computer-readable media of claim 19, wherein training the image comparison model comprises training the image comparison model to perform a structural property comparison between telemetry state images to identify whether or not there is a match between the telemetry state images.
